

Basic Applications/ Algorithms of

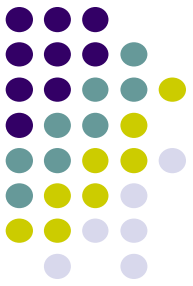
Morphological Image Processing

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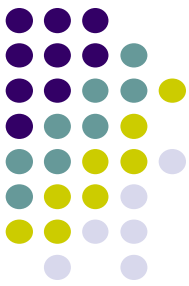
Boundary Extraction

$$\beta(A) = A - (A \ominus B)$$

- The boundary of a set A is obtained by first eroding A by structuring element B and then taking the set difference of A and its erosion.
- The resultant image has the boundary of the objects extracted.
- The thickness of the boundary depends on the size of the structuring element.



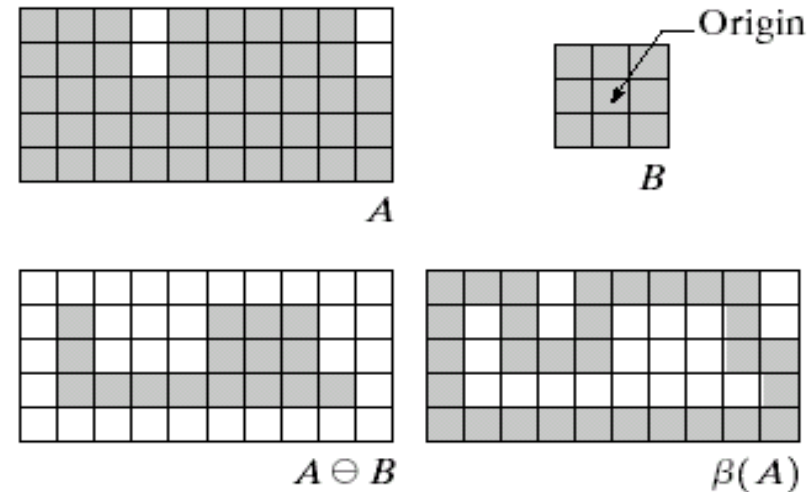
Boundary Extraction Example 1:



$$\beta(A) = A - (A \ominus B)$$

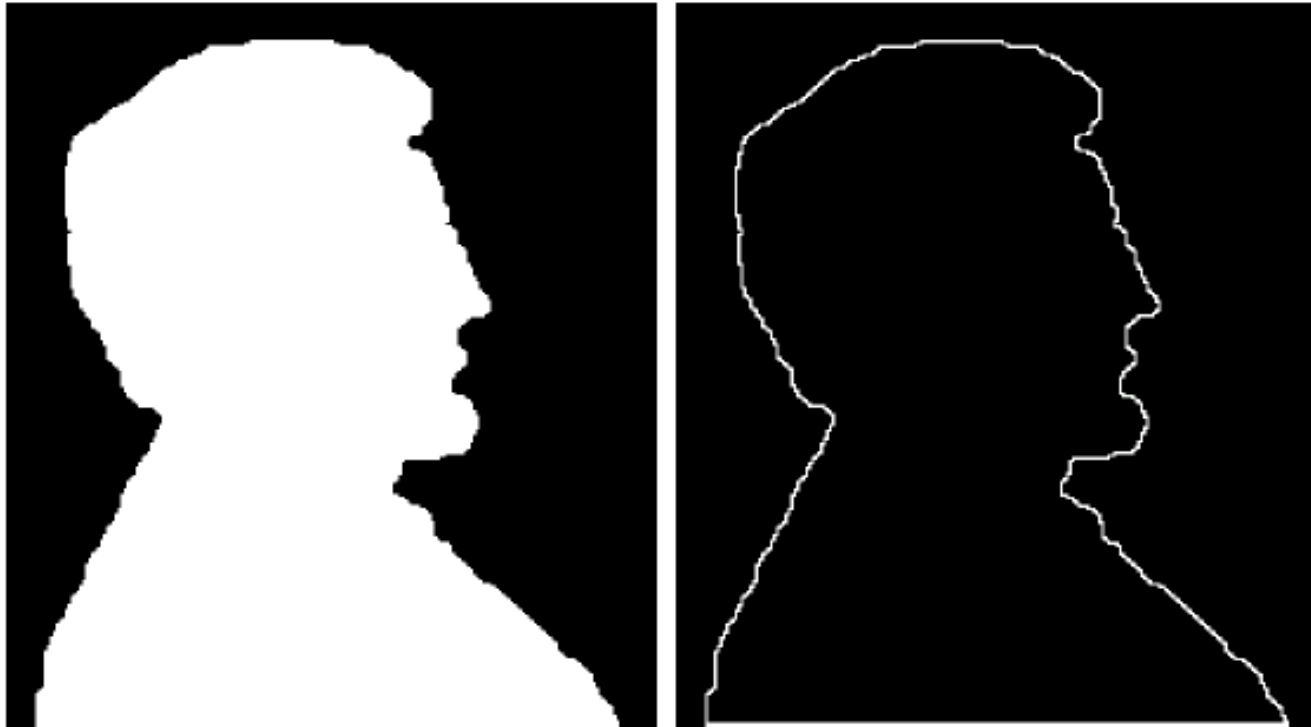
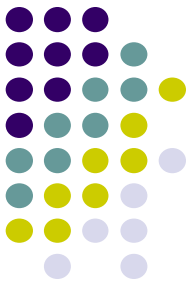
a	b
c	d

FIGURE 9.13 (a) Set A . (b) Structuring element B . (c) A eroded by B . (d) Boundary, given by the set difference between A and its erosion.





Boundary Extraction Example 2:



a b

FIGURE 9.14

(a) A simple binary image, with 1's represented in white. (b) Result of using Eq. (9.5-1) with the structuring element in Fig. 9.13(b).



Boundary Extraction Example 3:

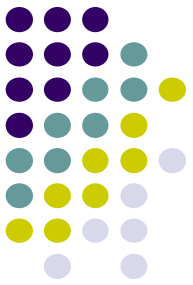
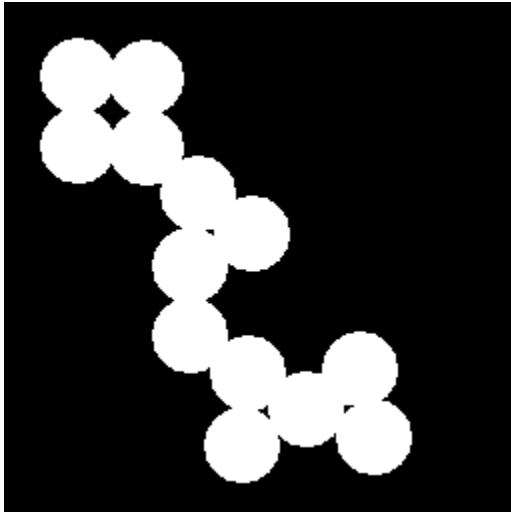
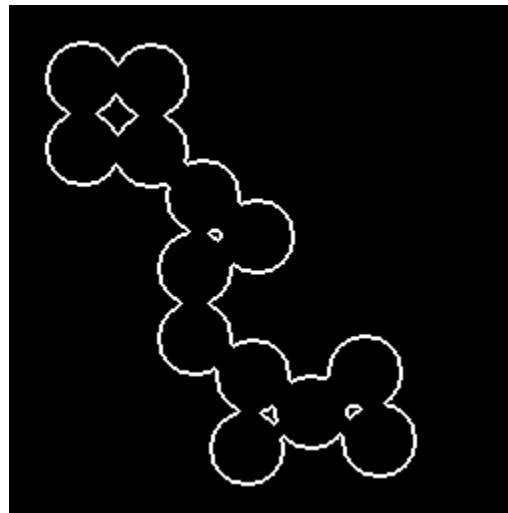


Image: Circles.png



Circles.png Boundaries



Matlab Code:

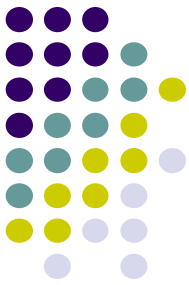
```
x = imread('circles.png');  
imshow(x)
```

```
% Structuring element.  
se = strel('disk',2);
```

```
y = imerode(x,se);  
z = imsubtract(x,y);  
figure, imshow(z)
```



Region Filling



- Start from p inside boundary.

$$X_0 = p$$

$$X_k = (X_k \oplus B) \cap A^c$$

$$\text{Until: } X_{k+1} = X_k$$

a	b	c
d	e	f
g	h	i

FIGURE 9.15

Region filling.

(a) Set A .

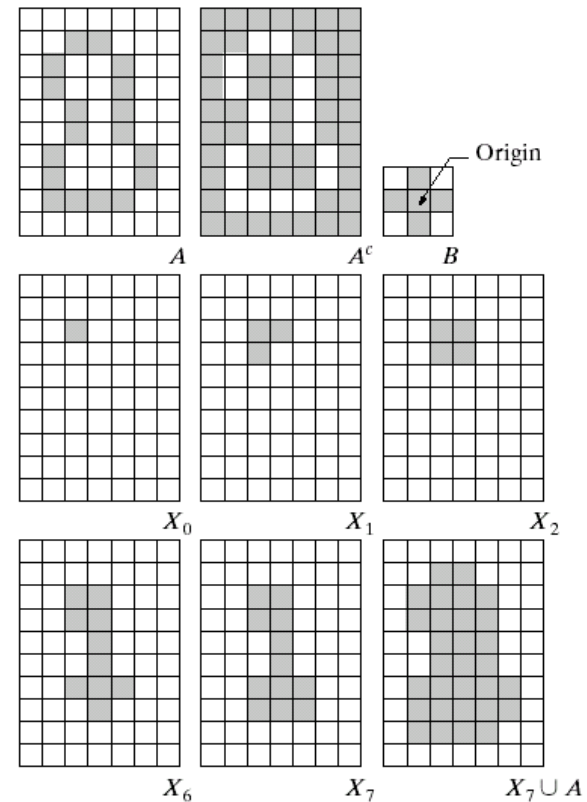
(b) Complement of A .

(c) Structuring element B .

(d) Initial point inside the boundary.

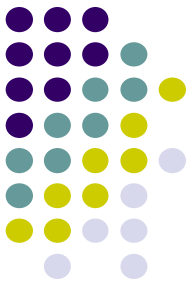
(e)–(h) Various steps of Eq. (9.5-2).

(i) Final result [union of (a) and (h)].

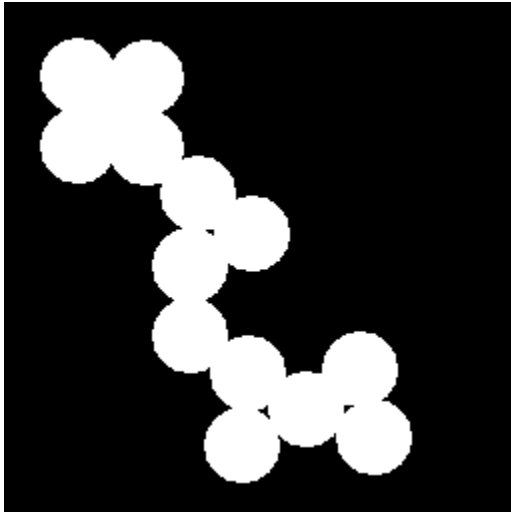




Region Filling Example 1



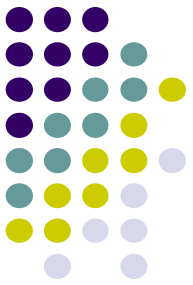
Region Filled of circle.png



```
A = imread('circles.png');  
imshow(A)  
se = strel('disk',2);
```

```
Xk = zeros(size(A));  
Xk1 = Xk;  
Xk(55, 54) = 1;  
while any(Xk(:) ~= Xk1(:))  
  Xk1 = Xk;  
  Xk = imdilate(Xk1, se) & ~A;  
end  
A1 = Xk | A;
```

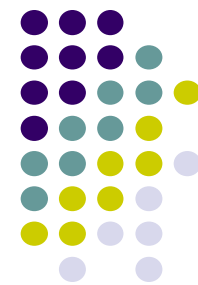
```
figure, imshow(A1)
```



Important Note on Matlab

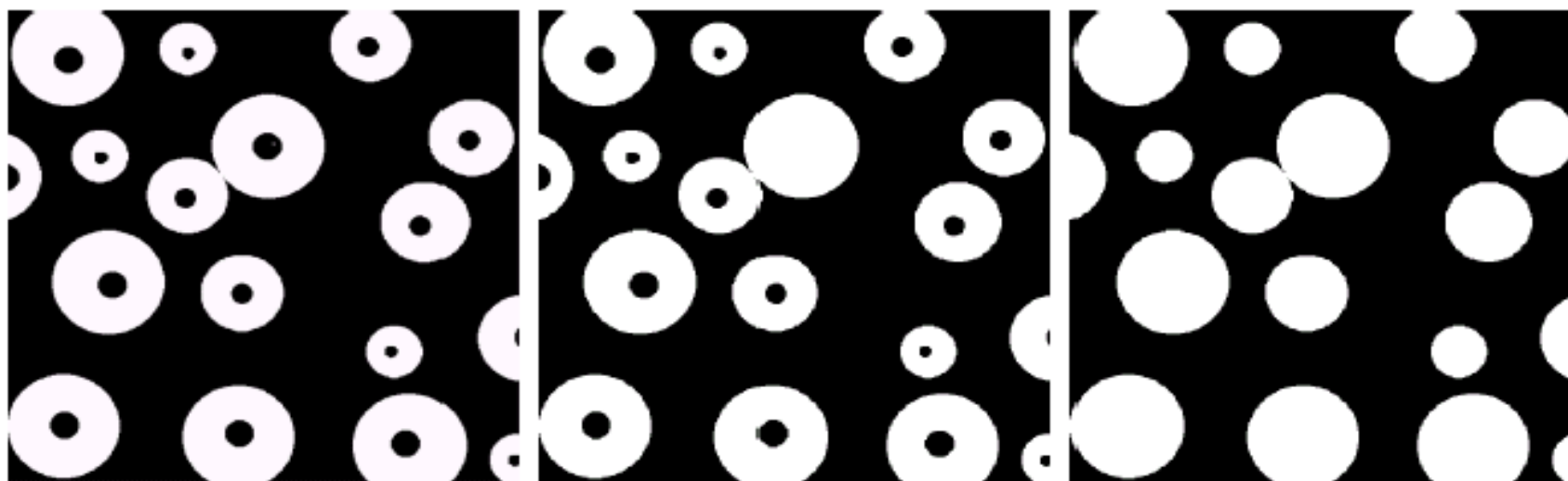
- MATLAB set operations on binary images

Set Operation	MATLAB Expression	Name
$A \cap B$	$A \& B$	AND
$A \cup B$	$A B$	OR
A^c	$\sim A$	NOT
$A - B$	$A \& \sim B$	DIFFERENCE



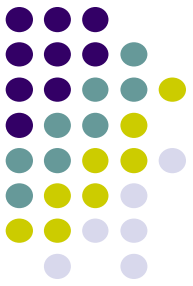
Region Filling Example 2

Semi-automated to cancel reflection effect



a b c

FIGURE 9.16 (a) Binary image (the white dot inside one of the regions is the starting point for the region-filling algorithm). (b) Result of filling that region (c) Result of filling all regions.



Problem with Region Filling Algorithm?

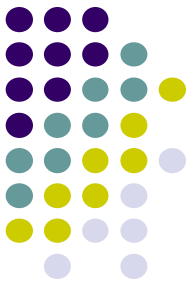
- Yes!:

You need to find the Initial Point!

Any Solution?



Connected components Extraction



Start from p belong to desired region.

$$X_0 = p$$

$$X_k = (X_k \oplus B) \cap A$$

Until: $X_{k+1} = X_k$

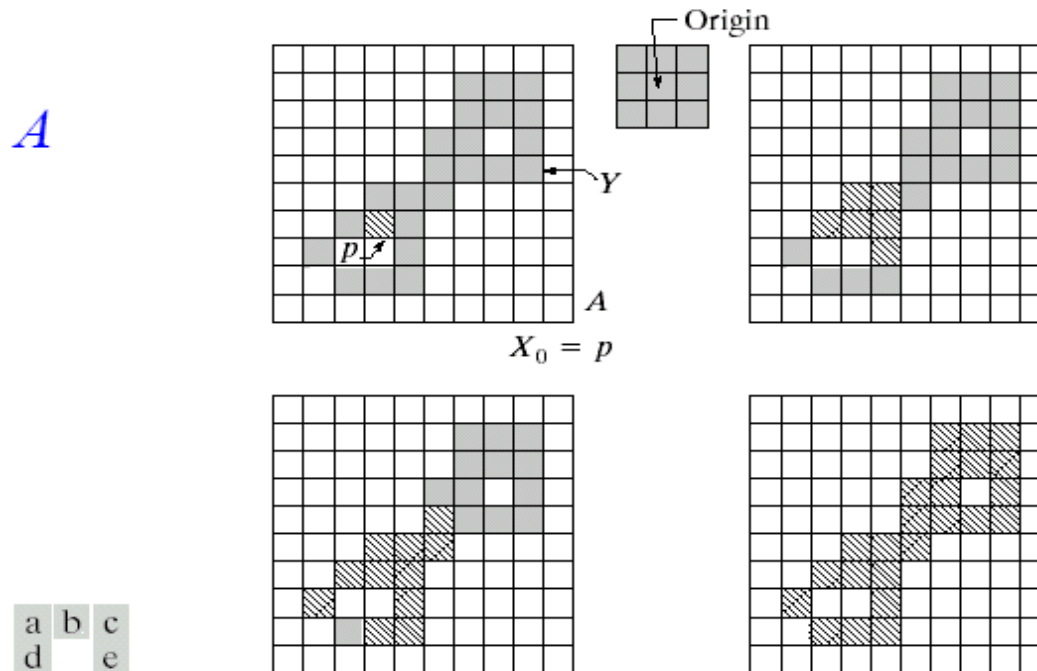
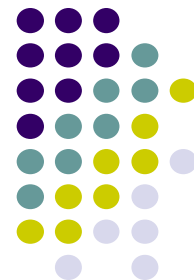


FIGURE 9.17 (a) Set A showing initial point p (all shaded points are valued 1, but are shown different from p to indicate that they have not yet been found by the algorithm). (b) Structuring element. (c) Result of first iterative step. (d) Result of second step. (e) Final result.



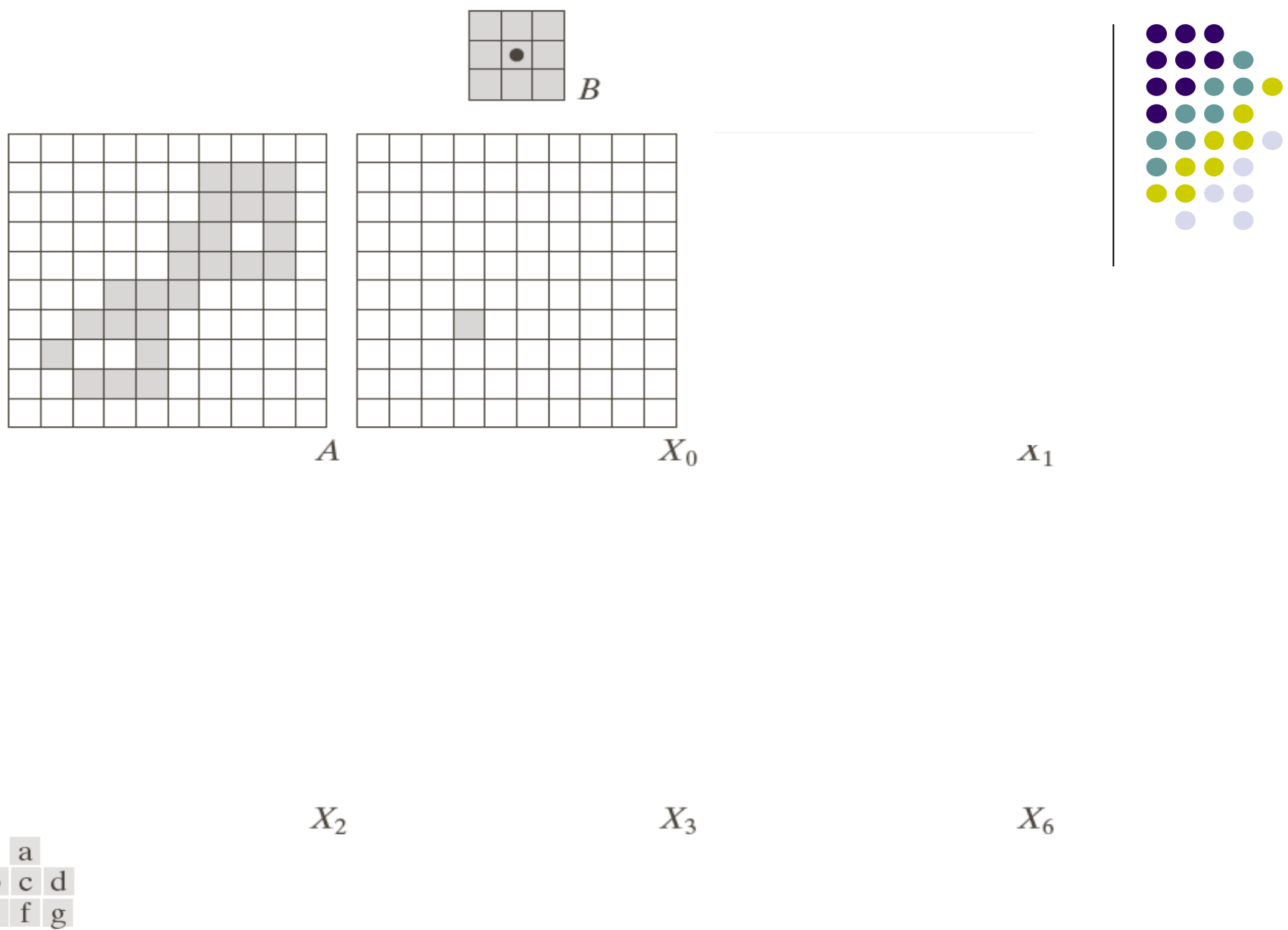
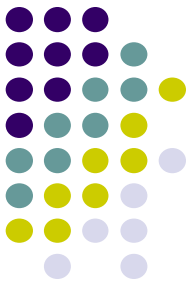


FIGURE 9.17 Extracting connected components. (a) Structuring element. (b) Array containing a set with one connected component. (c) Initial array containing a 1 in the region of the connected component. (d)–(g) Various steps in the iteration of Eq. (9.5-3).



CCE : Example 1



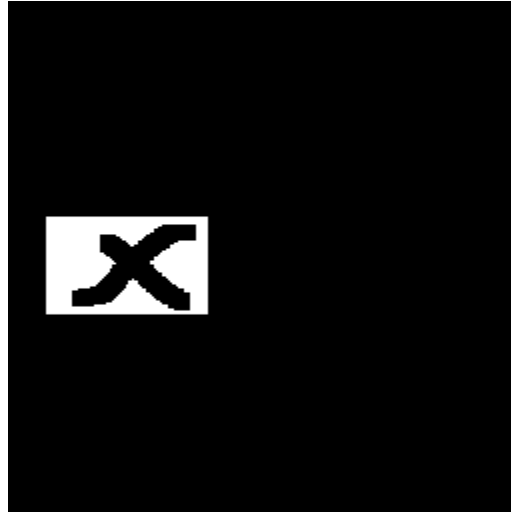
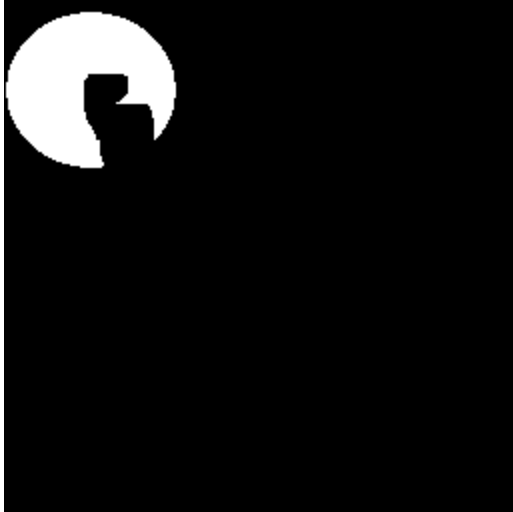
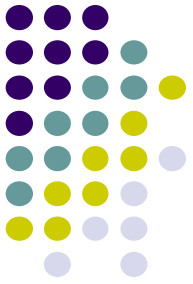
```
Xk = zeros(size(A));  
Xk1 = Xk;
```

```
% Xk(37, 26) = 1;  
% Xk(141, 91) = 1;  
Xk(224, 195) = 1;
```

```
while any(Xk(:) ~= Xk1(:))  
    Xk1 = Xk;  
    Xk = imdilate(Xk1, se) & A;  
end  
A1 = Xk;
```

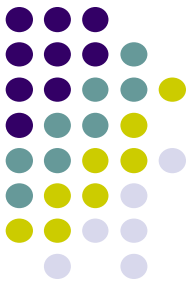


Results:





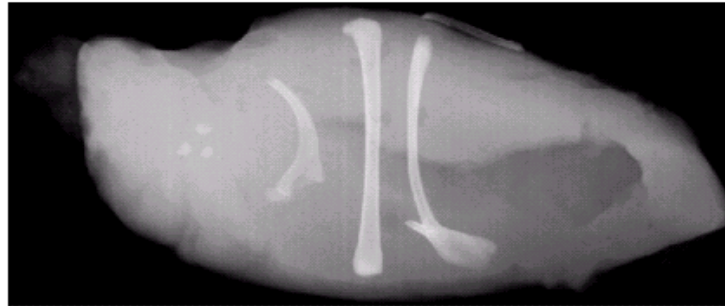
CCE Example 2



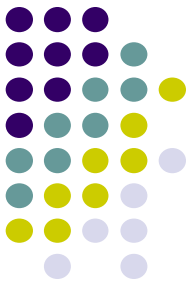
a
b
c d

FIGURE 9.18

(a) X-ray image of chicken filet with bone fragments.
(b) Thresholded image. (c) Image eroded with a 5×5 structuring element of 1's.
(d) Number of pixels in the connected components of (c). (Image courtesy of NTB Elektronische Geraete GmbH, Diepholz, Germany, www.ntbxray.com.)

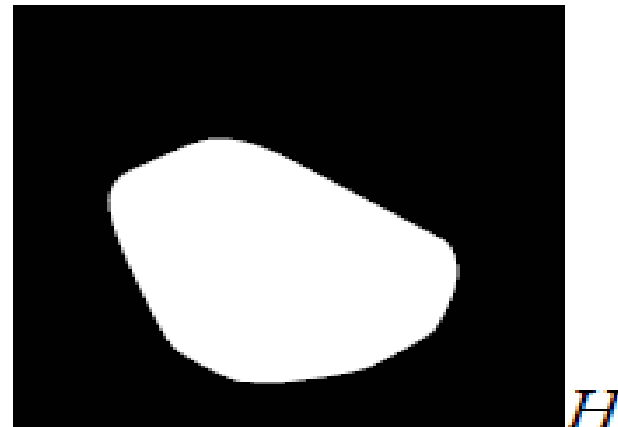
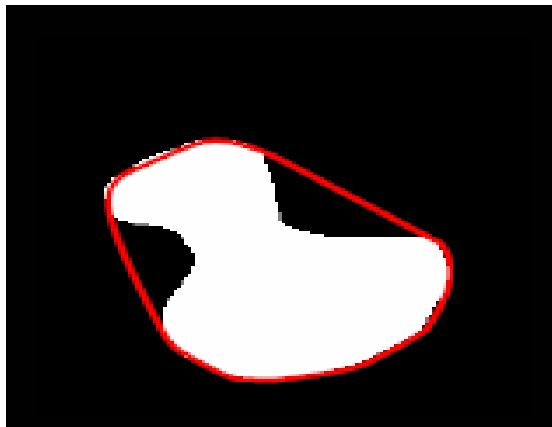


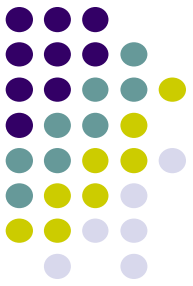
Connected component	No. of pixels in connected comp
01	11
02	9
03	9
04	39
05	133
06	1
07	1
08	743
09	7
10	11
11	11
12	9
13	9
14	674
15	85



Convex Hull

- A set A is said to be *convex* if the straight line segment joining any two points in A lies entirely within A .
- The *convex hull* H of an arbitrary set S is the smallest convex set containing S .





Convex Hull Continued:

- **Convex Hull Algorithm:**

Let B^i , $i = 1, 2, 3, 4$, represent the four structuring elements.

The procedure consists of implementing the equation:

$$X_k^i = (X_{k-1} \circledast B^i) \cup A$$

$$i = 1, 2, 3, 4 \text{ and } k = 1, 2, 3, \dots$$

with $X_0^i = A$.

When the procedure converges, or $X_k^i = X_{k-1}^i$, let $D^i = X_k^i$,
the convex hull of A is

$$C(A) = \bigcup_{i=1}^4 D^i$$

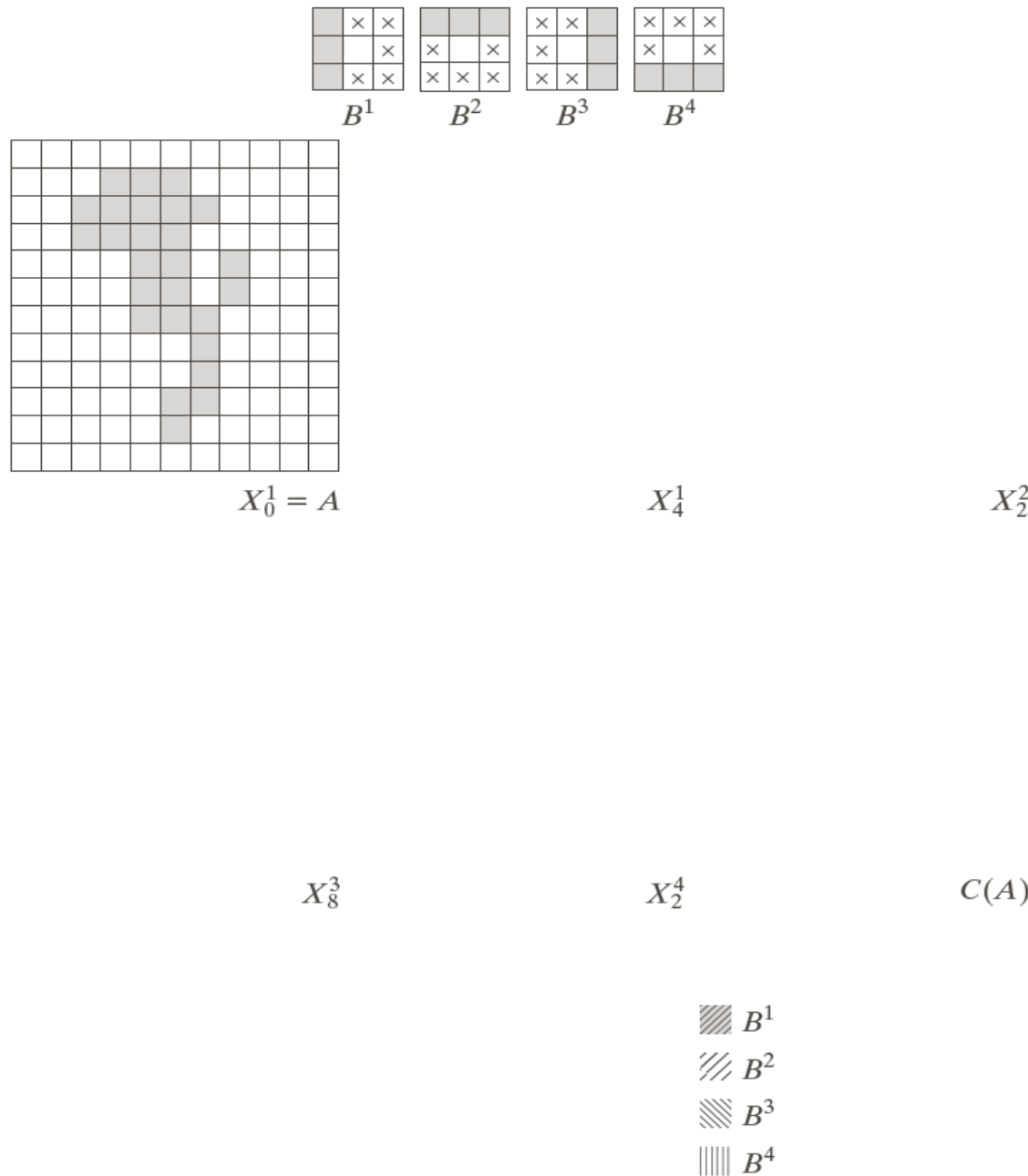
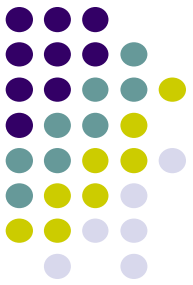


FIGURE 9.19
 (a) Structuring elements. (b) Set A . (c)–(f) Results of convergence with the structuring elements shown in (a). (g) Convex hull. (h) Convex hull showing the contribution of each structuring element.



The result:

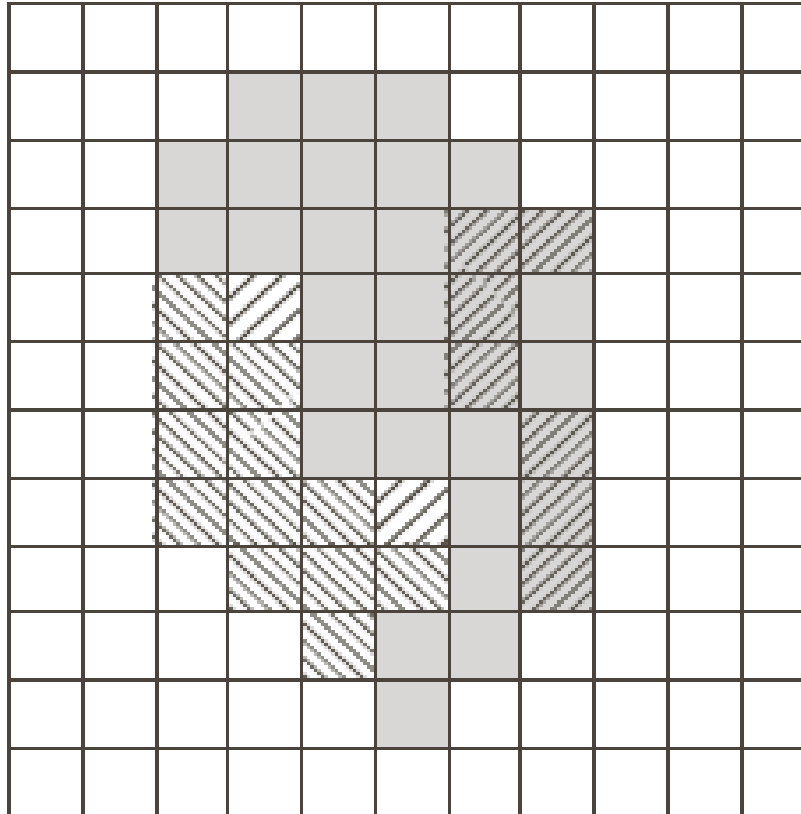
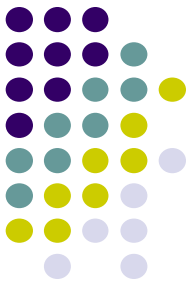


FIGURE 9.20
Result of limiting growth of the convex hull algorithm to the maximum dimensions of the original set of points along the vertical and horizontal directions.



Convex Hull Example 1



Original



Binary



Union Convex Hull



Objects Convex Hull



```
subplot(2,2,1);  
IC = imread('art1bw.bmp');  
I = imcomplement(IC);  
imshow(I); title('Original');
```

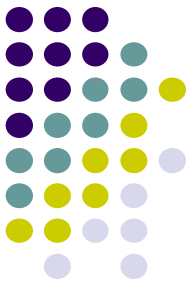
```
subplot(2,2,2);  
BW = I > 100;  
imshow(BW); title('Binary');
```

```
subplot(2,2,3);  
CH = bwconvhull(BW);  
imshow(CH); title('Union Convex Hull');
```

```
subplot(2,2,4);  
CH_O = bwconvhull(BW,'objects');  
imshow(CH_O); title('Objects Convex  
Hull');
```



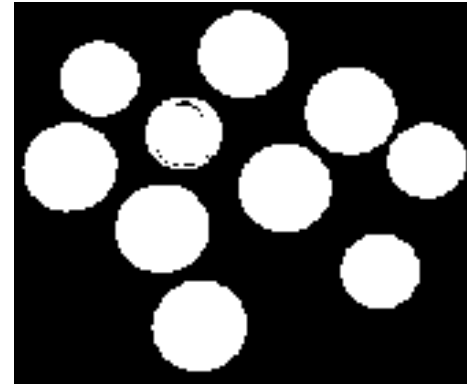
Convex Hull Example 2



Original



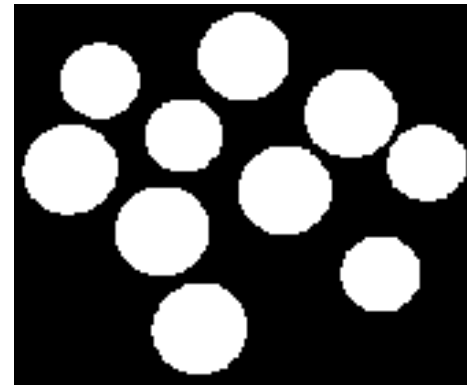
Binary

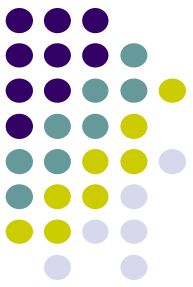


Union Convex Hull



Objects Convex Hull





Thinning

- Thinning Algorithm:

$$A \otimes B = A - (A * B) = A \cap (A * B)^c$$

Another approach:

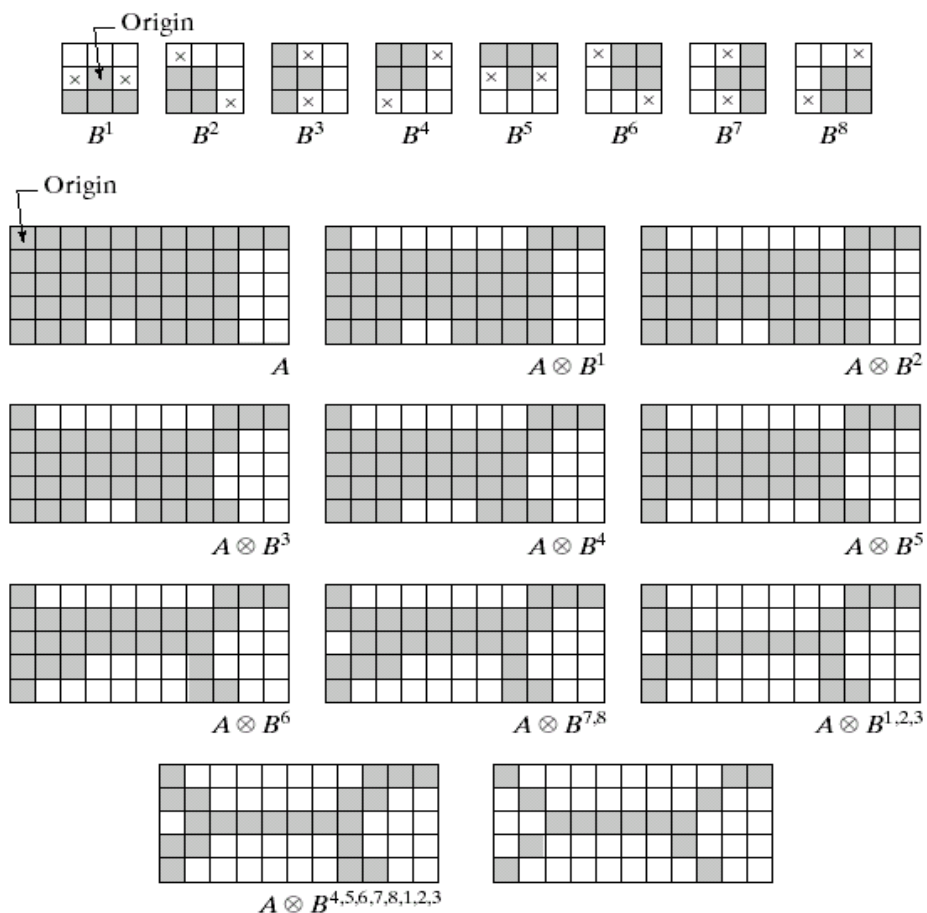
$$\{\mathbf{B}\} = \{B^1, B^2, \dots, B^n\}$$

$$A \otimes \{\mathbf{B}\} = \left(\left(\dots \left((A \otimes B^1) \otimes B^2 \right) \dots \right) \otimes B^n \right)$$

Repeat until convergence

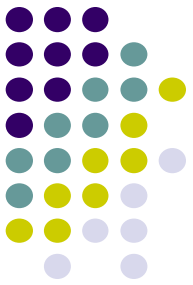


Thinning Cont.



a
b c d
e f g
h i j
k l

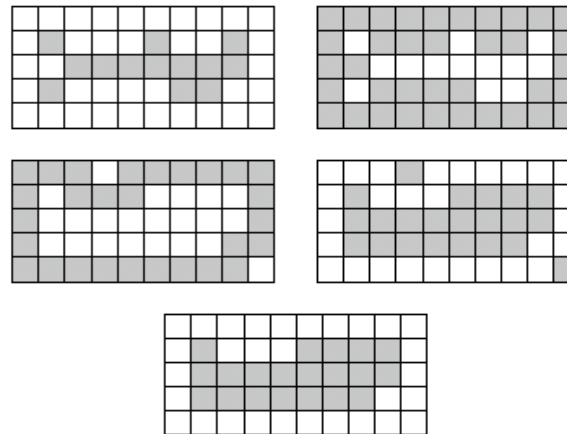
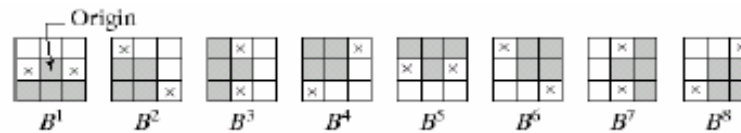
FIGURE 9.21 (a) Sequence of rotated structuring elements used for thinning. (b) Set A . (c) Result of thinning with the first element. (d)–(i) Results of thinning with the next seven elements (there was no change between the seventh and eighth elements). (j) Result of using the first element again (there were no changes for the next two elements). (k) Result after convergence. (l) Conversion to m -connectivity.



Thickening

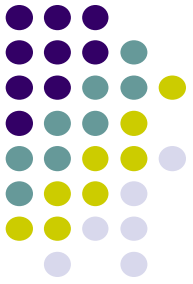
$$A \odot B = A \cup (A * B)$$

$$A \odot \{\mathbf{B}\} = \left(\left(\left(\left(A \odot B^1 \right) \odot B^2 \right) \cdots \right) \odot B^n \right)$$



a b
c d
e

FIGURE 9.22 (a) Set A . (b) Complement of A . (c) Result of thinning the complement of A . (d) Thickened set obtained by complementing (c). (e) Final result, with no disconnected points.

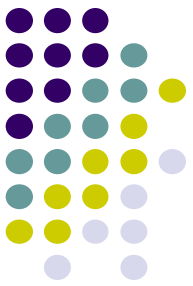


Thickening

- A with notation $S(A)$:
 - For z belong to $S(A)$ and $(D)z$, the largest disk
 - centered at z and contained in A , one can not find a
 - larger disk containing $(D)z$ and included in A .
 - Disk $(D)z$ touches the boundary of A at two or
 - more different points.



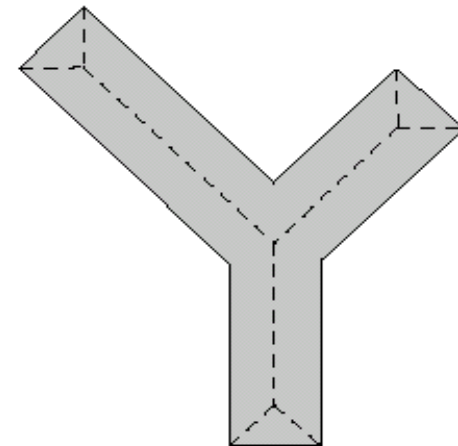
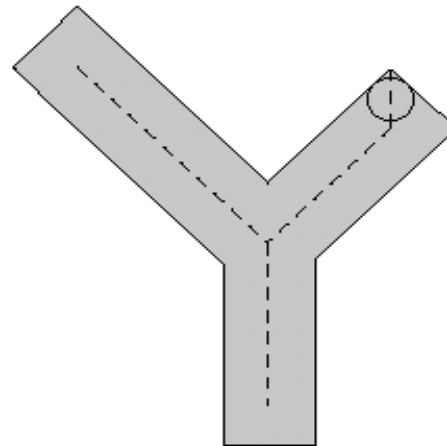
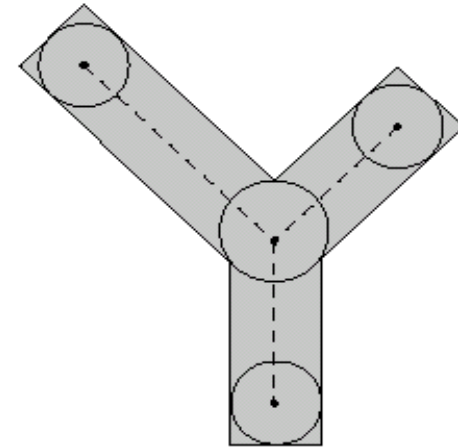
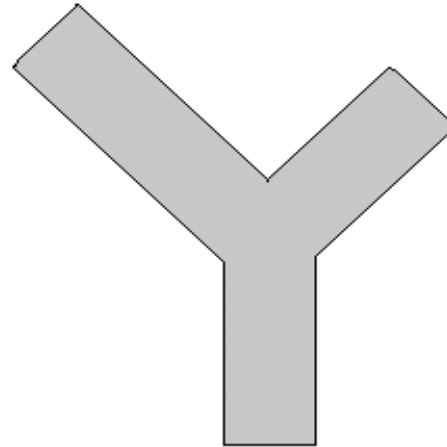
Skeletonization

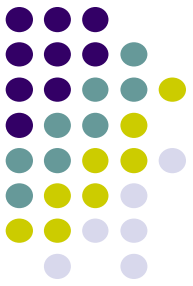


a	b
c	d

FIGURE 9.23

- (a) Set A .
(b) Various positions of maximum disks with centers on the skeleton of A .
(c) Another maximum disk on a different segment of the skeleton of A .
(d) Complete skeleton.





□ Skeletons A with notation $S(A)$:

$$S(A) = \bigcup_{k=0}^K S_k(A)$$

$$S_k(A) = (A \ominus kB) - (A \ominus kB) \circ B, \quad \circ: \text{Opening}$$

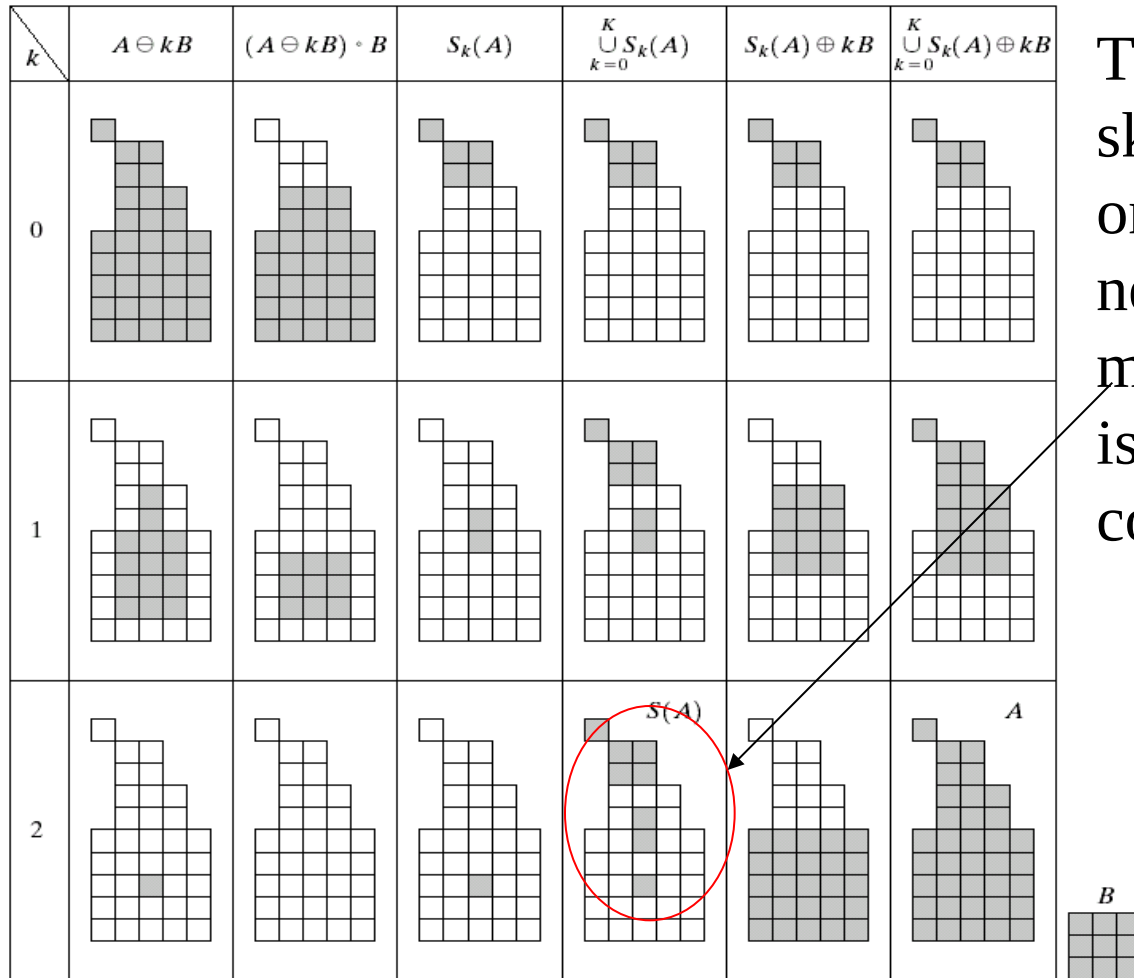
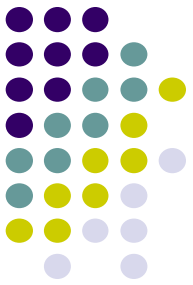
$$(A \ominus kB) = \left(\left(\dots \left((A \ominus B) \ominus B \right) \dots \right) \ominus B \right): \quad k \text{ times}$$

$$K = \max \{k \mid (A \ominus kB) \neq \emptyset\}$$

Reconstruction:

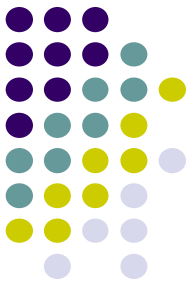
$$A = \bigcup_{k=0}^K (S_k(A) \oplus kB)$$

$$(A \oplus kB) = \left(\left(\dots \left((A \oplus B) \oplus B \right) \dots \right) \oplus B \right): \quad k \text{ times}$$



The final skeleton not only thicker it needs to be ,but more important is not connected.

FIGURE 9.24 Implementation of Eqs. (9.5-11) through (9.5-15). The original set is at the top left, and its morphological skeleton is at the bottom of the fourth column. The reconstructed set is at the bottom of the sixth column.



Pruning

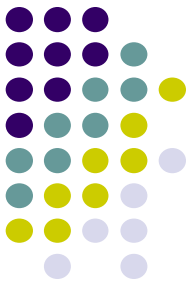
Removing parasitic component complement to thinning and skeletonizing by successive elimination its end point.

$$X_1 = A \otimes \{B\}$$

$$X_2 = \bigcup_{k=1}^8 (X_1 \circledast B^k)$$

$$X_3 = (X_2 \oplus H) \cap A$$

$$X_4 = X_1 \cup X_3$$



a b
c
d e
f g

FIGURE 9.25

(a) Original image. (b) and (c) Structuring elements used for deleting end points. (d) Result of three cycles of thinning. (e) End points of (d). (f) Dilation of end points conditioned on (a). (g) Pruned image.

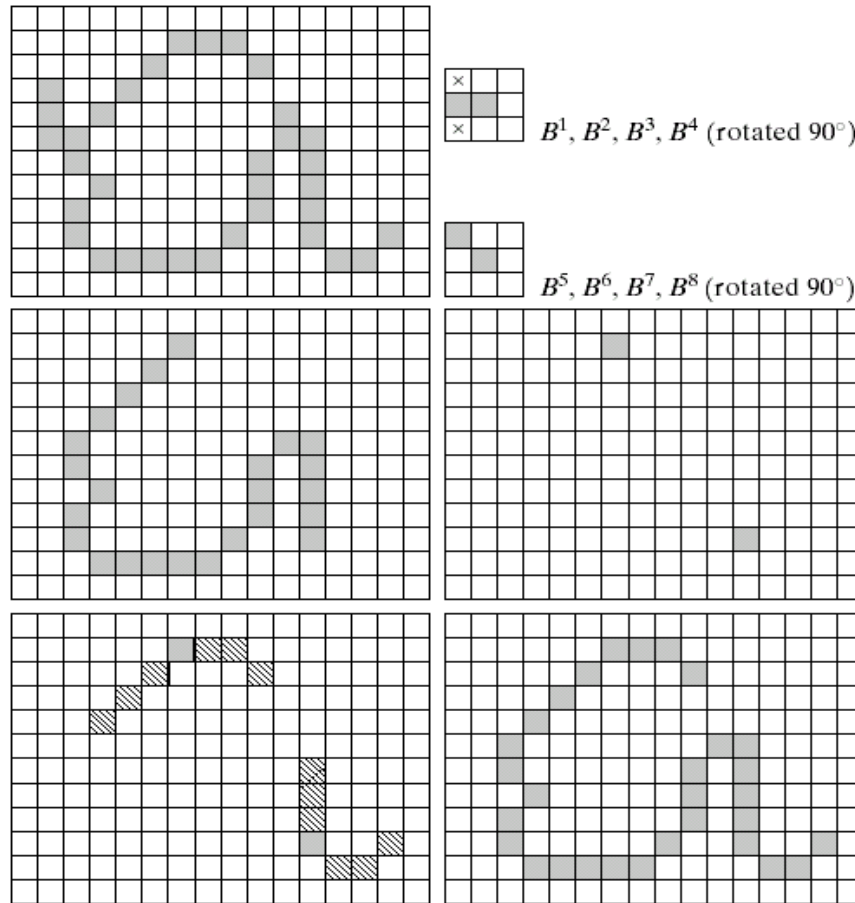
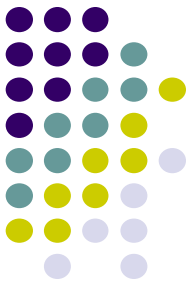


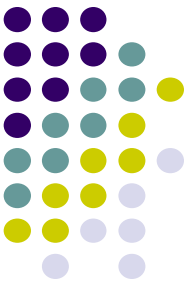


TABLE 9.2

Summary of morphological operations and their properties.

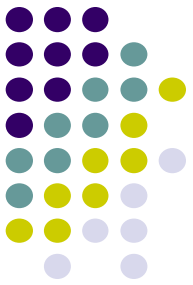
Operation		Equation	Comments (The Roman numerals refer to the structuring elements shown in Fig. 9.26).
Translation		$(A)_z = \{w w = a + z, \text{ for } a \in A\}$	Translates the origin of A to point z .
Reflection		$\hat{B} = \{w w = -b, \text{ for } b \in B\}$	Reflects all elements of B about the origin of this set.
Complement		$A^c = \{w w \notin A\}$	Set of points not in A .
Difference		$A - B = \{w w \in A, w \notin B\}$ $= A \cap B^c$	Set of points that belong to A but not to B .
Dilation		$A \oplus B = \{z (\hat{B})_z \cap A \neq \emptyset\}$	“Expands” the boundary of A . (I)
Erosion		$A \ominus B = \{z (B)_z \subseteq A\}$	“Contracts” the boundary of A . (I)
Opening		$A \circ B = (A \ominus B) \oplus B$	Smooths contours, breaks narrow isthmuses, and eliminates small islands and sharp peaks. (I)
Closing		$A \bullet B = (A \oplus B) \ominus B$	Smooths contours, fuses narrow breaks and long thin gulfs, and eliminates small holes. (I)





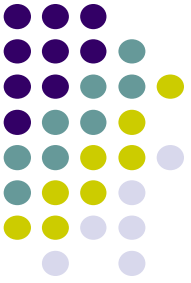
Hit-or-miss transform	$A \circledast B = (A \ominus B_1) \cap (A^c \ominus B_2)$ $= (A \ominus B_1) - (A \oplus \hat{B}_2)$	The set of points (coordinates) at which, simultaneously, B_1 found a match ("hit") in A and B_2 found a match in A^c .
Boundary extraction	$\beta(A) = A - (A \ominus B)$	Set of points on the boundary of set A . (I)
Region filling	$X_k = (X_{k-1} \oplus B) \cap A^c; X_0 = p \text{ and } k = 1, 2, 3, \dots$	Fills a region in A , given a point p in the region. (II)
Connected components	$X_k = (X_{k-1} \oplus B) \cap A; X_0 = p \text{ and } k = 1, 2, 3, \dots$	Finds a connected component Y in A , given a point p in Y . (I)
Convex hull	$X_k^i = (X_{k-1}^i \circledast B^i) \cup A; i = 1, 2, 3, 4;$ $k = 1, 2, 3, \dots; X_0^i = A; \text{ and}$ $D^i = X_{\text{conv}}^i.$	Finds the convex hull $C(A)$ of set A , where "conv" indicates convergence in the sense that $X_k^i = X_{k-1}^i$. (III)

TABLE 9.2
Summary of morphological results and their properties.
(continued)



Operation		Equation	Comments (The Roman numerals refer to the structuring elements shown in Fig. 9.26).
Thinning		$A \otimes B = A - (A \otimes B)$ $= A \cap (A \otimes B)^c$	Thins set A. The first two equations give the basic definition of thinning. The last two equations denote thinning by a sequence of structuring elements. This method is normally used in practice. (IV)
		$A \otimes \{B\} =$ $((\dots ((A \otimes B^1) \otimes B^2) \dots) \otimes B^n)$ $\{B\} = \{B^1, B^2, B^3, \dots, B^n\}$	
Thickening		$A \odot B = A \cup (A \otimes B)$	Thickens set A. (See preceding comments on sequences of structuring elements.) Uses IV with 0's and 1's reversed.
		$A \odot \{B\} =$ $((\dots (A \odot B^1) \odot B^2 \dots) \odot B^n)$	

TABLE 9.2
Summary of morphological results and their properties.
(*continued*)



Thank you

for your attention